For an **Additive White Gaussian Noise (AWGN)** channel, the received signal is

$$Y = X + N$$

where:

- X= transmitted signal
- N= Gaussian noise with

$$N \sim \mathcal{N}(0, N_0/2)$$

(or variance σ^2)

- Y= received signal

The transmitter is constrained by an average power:

$$E[X^2] \leq P$$

We derive the **channel capacity** C, which is the maximum reliable transmission rate.

Step 1: Start from Shannon Capacity Definition

Channel capacity is

$$C = \max_{p(x)} I(X;Y)$$

where $I(X;Y)$is the mutual information.

How much information the received signal Ycontains about the transmitted signal X.

X = transmitted signal

Y = received signal

the term p(x) means: The probability distribution of the transmitted signal X.

The “max” means: We choose the best possible input distribution p(x) that maximizes the mutual information.

Using the definition:

$$I(X;Y) = h(Y) - h(Y \mid X)$$

h(Y) is the differential entropy of the continuous random variable Y.

h(Y|X) is the conditional differential entropy of Y given X. It measures: The remaining uncertainty in the received signal Y when the transmitted signal X is already known.

For AWGN:

$$Y = X + N$$

Since N is independent of X,

$$h(Y \mid X) = h(N)$$

Thus,

$$C = \max_{p(x)} [h(Y) - h(N)]$$

Step 2: Entropy of Gaussian Noise

If noise variance is σ^2,

$$N \sim \mathcal{N}(0, \sigma^2)$$

then the differential entropy is

$$h(N) = \frac{1}{2} \log_2 (2\pi e \sigma^2)$$

Step 3: Maximize $h(Y)$

Since

$$Y = X + N$$

the variance of $Y$is

$$\mathrm{Var}(Y) = P + \sigma^2$$

For a fixed variance, the **Gaussian distribution gives maximum entropy**.

Therefore, capacity is achieved when:

$$X \sim \mathcal{N}(0, P)$$

Then $Y$is also Gaussian:

$$Y \sim \mathcal{N}(0, P + \sigma^2)$$

Hence,

$$h(Y) = \frac{1}{2}\log_2(2\pi e(P + \sigma^2))$$

Step 4: Compute Mutual Information

Substitute into

$$C = h(Y) - h(N)$$
$$C = \frac{1}{2}\log_2(2\pi e(P + \sigma^2)) - \frac{1}{2}\log_2(2\pi e\sigma^2)$$

Using log rules:

$$C = \frac{1}{2}\log_2\left(\frac{P + \sigma^2}{\sigma^2}\right)$$
$$C = \frac{1}{2}\log_2\left(1 + \frac{P}{\sigma^2}\right)$$

Final Channel Capacity Formula

Capacity per channel use:

$$\boxed{C = \frac{1}{2}\log_2\left(1 + \frac{P}{\sigma^2}\right) \text{ bits/channel use}}$$

If bandwidth BHz is considered, then

Because a bandwidth-B channel can carry at most 2B independent real samples per second. Also, for AWGN, N~N(0,σ^2), s, mean =0
Also, for Noise Power = $N = E\{x^2\} - (E\{x\})^2 = \sigma^2 - \mu = \sigma^2 - 0 = \sigma^2$

$$\boxed{C = B\log_2\left(1 + \frac{P}{N}\right) \text{ bits/s}}$$

where

- P= signal power
- N= noise power
- P/N= SNR

Important Notes

1. Capacity Depends on SNR

$$\text{SNR} = \frac{P}{N}$$

Higher SNR ⇒ higher capacity.

2. Gaussian Input Maximizes Capacity

The optimal input distribution is Gaussian:

$$X \sim \mathcal{N}(0, P)$$

3. Shannon-Hartley Theorem

The AWGN channel capacity formula is also known as the **Shannon-Hartley theorem**.

Compact Derivation Summary

$$\begin{aligned} C &= \max I(X;Y) \\ &= \max [h(Y) - h(Y \mid X)] \\ &= \max [h(Y) - h(N)] \\ &= \frac{1}{2}\log_2(2\pi e(P+\sigma^2)) - \frac{1}{2}\log_2(2\pi e\sigma^2) \\ &= \frac{1}{2}\log_2\left(1+\frac{P}{\sigma^2}\right) \end{aligned}$$

The key graphable relationship is:

$$C = \frac{1}{2}\log_2\left(1+\frac{P}{\sigma^2}\right)$$